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Theory and Methodology

The hub location and routing problem

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Abstract

In this paper, we consider the hub location and routing problem in which the hub locations and the service types for the routes between demand points are determined together. Rather than aggregating the demand for the services, flows from an origin to different destination points are considered separately. For each origin–destination pair, one-hub-stop, two-hub-stop and, when permitted, direct services are considered. In the system considered, the hubs interact with each other and the level of interaction between them is determined by the two-hub-stop service routes. A mathematical formulation of the problem and an algorithm solving the hub location and the routing subproblems separately in an iterative manner are presented. Computational experience with four versions of the proposed algorithm differing in the method used for finding starting solutions is reported.

Keywords: Location; Distribution; Heuristics

1. Introduction

Although several studied involving location of new facilities with fixed amounts of interfacility flows have appeared in the literature (e.g. multifacility location problem, p-median and p-center problems with mutual communication, Kolen, 1986, and Chhajed and Lowe, 1992), previous research on the location-allocation problem was focused primarily on the situations involving location of non-interacting new facilities. It is usually assumed that new facilities would not interact with other new facilities nor would they interact with the demand points assigned to other new facilities. This assumption of the location-allocation problem has been relaxed recently in the studies on the hub location problems on the plane E^2 (O'Kelly, 1986; Aykin, 1988). In a system employing hubs, item flow occurs between demand points through a set of new facilities (hubs) acting as switching/concentration points. Flows between two demand points, say i and j, are either sent through one hub, if both i and j are served by the same hub (that is, from i to hub and from hub to j), or through two hubs, if they are served by different hubs (that is, from i to hub serving i, from hub serving i to hub serving j and finally from hub serving j to j). Thus, besides the interactions between the hub facilities and demand points considered in the location–allocation problem, interactions between the hubs themselves exist due to the flows channelled between them. These inter-hub flows affect both the location and the allocation decisions. The hub location problem on the plane E^2 with one and two hubs was first studied by O'Kelly (1986) and with three hubs

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by Fotheringham and O'Kelly (1989). A formulation of the p-hub location problem is given in Aykin (1988) and a heuristic procedure using a flow based allocation rule in Aykin and Brown (1992).

In discrete solution space, a quadratic integer programming formulation of the p-hub location problem was provided by O'Kelly (1987). He proposed two proximity based heuristics. Use of proximity based heuristics for this problem is also discussed in Aykin (1990) together with a flow based allocation rule. A number of exact and heuristic approaches have been proposed for this problem recently: A number of interchange and covering heuristics by Klincewicz (1991), a branch and bound procedure and a simulated annealing based interchange heuristic by Aykin (1991), two heuristics based on the solution of the problem with relaxed routing (relaxing the single-hub-assignment restriction for the demand points) by Campbell (1991) and heuristics based on tabu search by Skorin-Kapov and Skorin-Kapov (1992) and Klincewicz (1993).

Another assumption made in the studies on the location-allocation problem is that each demand point is best served if it is assigned to *one* new facility for service. This is also a common assumption in the studies involving interacting hub facilities. This assumption may be appropriate if the demand for services is homogeneous and can be provided from any of the new facilities. In a system employing hubs, however, directional demand exists at the demand points for the services provided between them. Furthermore, due to economies of scale in the increased flows between the hubs, the unit transportation costs between the hubs are expected to be lower than the unit transportation costs between a demand point and a hub, and between two demand points. To minimize transportation costs in this case, flows from a demand point to different destinations should be considered separately and the most economical service type should be determined for each route (identified by its origin and destination) originating from that demand point. Consequently, a demand point may be served by different hubs on different routes.

One of the successful applications of the hubbing concept is found in air passenger/cargo transportation. In air transportation, air travel hubs act as the switching points channeling flows between the cities. To improve service in air passenger transportation, more convenient nonstop flights are also offered by the airlines between some non-hub cities. For each route, a decision regarding the service type (Fig. 1) is made; flows between two cities are either shipped nonstop (direct shipping with no hub-stop) or through hub(s) (one- or more-hub-stop). Federal Express, for instance, has major hubs at Indianapolis, IN, Memphis, TN, Newark, NJ and Oakland, CA. A package from San Francisco may be shipped via Oakland, Indianapolis, Memphis, Newark or an appropriate combination of hubs depending on the destination. Although a policy requiring shipment of all flows to/from a city through the same hub (single hub assignment) has the advantages of network simplicity and possible higher facility utilization,

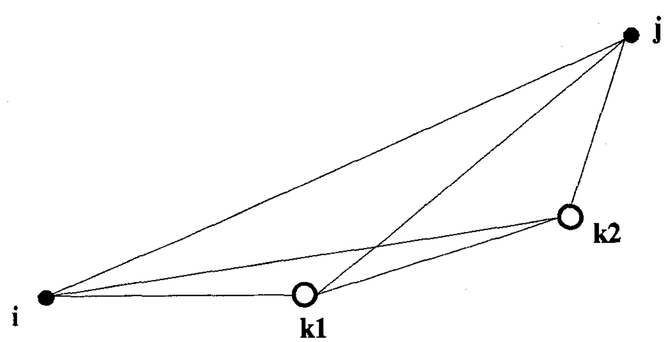

Fig. 1. Nonstop vs. hub connected service.

this may not be acceptable because of the operational restrictions it imposes on the system. Passengers, for example, may choose more convenient services rather than making one or more hub-stops or long detours every time they fly. In express cargo delivery, this type of network organization may require more time and resources for shipping items between some cities than the operating/marketing policies allow (e.g. guaranteed next day delivery, and, in some cases, delivery before a certain time in the morning).

Recently, a number of researchers have considered the routing problem in many-to-many (many origins and many destinations) freight networks with directional demand. Blumenfeld et al. (1984) studied the optimal shipping strategies for freight networks involving direct shipping and shipping via a consolidation terminal. Hall (1987) considered direct versus terminal routing for networks with a single consolidation terminal, large number of origins and few destinations. Hall (1989) studied the same problem with multiple consolidation terminals. Recently, Klincewicz (1990) presented an integer programming formulation of the freight transportation problem with multiple consolidation terminals and a piecewise linear concave shipping cost function. Assuming known terminal locations, he considered direct shipping and shipping via a consolidation terminal. A common assumption in these studies is that the locations of the service facilities (or consolidation terminals, distribution centers, as they are referred to in some studies) are known. It is known, however, that the routing decisions are affected by the locations of the service facilities and the locations of the service facilities by the routing decisions. Recognizing this relation, the hub location and the routing problems have recently been studied together. Aykin (1991) considered the discrete hub location and routing problem with nonstop, one-hub-stop and two-hub-stop services. He presented an integer programming formulation of this problem together with an enumeration algorithm and a drop and interchange heuristic. Campbell (1991) provided a similar formulation for this problem without nonstop services between non-hub demand points and used an enumeration algorithm to solve it. O'Kelly and Lao (1991) considered a hub-and-spoke system design problem involving two hubs with known locations (referred to as minihub and masterhub), only one-hub-stop service and choice of truck or air service for each route. They present a quadratic integer programming formulation for the service selection and routing problem in discrete solution space and solve it through linearization. Concerning the locations of hubs, they suggest a possible use of their approach in a procedure enumerating all pairs of potential hub locations. Aykin (1994) studied the capacitated hub location and routing problem in discrete solution space with nonstop, one-hub-stop and two-hub-stop services and constraints on hub capacities. He presented two Lagrangian relaxation based heuristics.

In this paper, we introduce the hub location and routing problem on the plane E^2. Specifically, we consider a system in which flows from one demand point to another are either channelled through hubs (one-hub-stop or two-hub-stop services) or, if permitted, shipped directly with no hub-stop. Depending on the application, nonstop services may be permitted for all, some or none of the routes served by the system. The set of routes for which direct shipping is permitted is assumed to be predetermined (an input to the problem). The problem is to determine the service types and hub locations on the plane that minimize the total transportation cost. This problem is an extension of the well known location–allocation problem with its assumptions concerning the interactions between new facilities and the allocation of demand relaxed. We present a formulation of the problem and propose a solution algorithm together with four methods for finding starting solutions. The model studied in this paper assumes that the hubs can be located anywhere in the service region. One of the areas where the analysis presented in this study can be applied is the design of hub-and-spoke systems involving super hubs. Recently, the development of super hubs has been proposed as one possible solution to the airport capacity shortage problem (Hamzawi, 1992). Since a large portion of the passengers flying to major hub airports stay there only few hours and change planes to fly to other airports, super hubs may be developed specifically for the connecting traffic as transfer airports. This way the connecting passengers can fly to super hubs, change planes and fly to other super hubs or to their destinations. To avoid the traffic congestion problems in airspace and on the ground, the super hubs should not be located at metropolitan areas but rather at rural locations within the service region (Hamzawi, 1992). Other areas of application for the analysis presented include retailing, emergency relief distribution and telecommunications (Brandon, 1987).

The rest of the paper is organized as follows: In Section 2, we present a formulation of the hub location and routing problem. To study the effects of nonstop services on the route structure and hub locations, we consider two limiting cases: (i) 100% case in which nonstop services are permitted between all demand points and (ii) 0% case in which nonstop services are not permitted between (non-hub) demand points. In Section 3, properties of the model are explored and an algorithm solving smaller subproblems in an iterative manner is presented. In Section 4, we describe four methods for finding starting solutions for the algorithm presented in Section 3. Computational experience is reported in Section 5.

2. Problem formulation

Let $P_i \in E^2$ be the vector location of demand point i, $i = 1, \dots, n$, and c_{ik} be the cost per unit distance of transporting one unit of flow directly from i to k. Let W_{ij} be the amount of flows from demand point i to demand point j. Denote the vector locations of p interacting hubs by $Q_k \in E^2$, $k = 1, \dots, p$. Let $d(P, Q)$ be the distance between the points P and Q. Let a_1, a, $a_2 \in [0, 1]$ be the proportionality constants quantifying scale economies for the segments of a hub-connected service route from its origin to a hub, between two hubs and from a hub to its destination, respectively, $a \leq a_1, a_2$. The cost per unit distance per unit of flow for one-stop and two-stop services is expected to be lower than that for nonstop service due to increased traffic intensity and centralization of the operations at hubs. The distance travelled with one-stop and two-stop services, however, is longer. Let H be the set of routes (i, j) (with origin i and destination j) for which nonstop service is permitted. Further, let O_{ij} be equal to one if $(i, j) \in H$ and zero otherwise. Define the following decision variables:

$$X_{ij} = \begin{cases} 1 & \text{if flows from } i \text{ to } j \text{ are shipped directly,} \\ 0 & \text{otherwise,} \end{cases}$$

and

$$X_{iktj} = \begin{cases} 1 & \text{if flows from } i \text{ to } j \text{ are shipped with the routing } i \to \text{hub } k \to \text{hub } t \to j. \\ 0 & \text{otherwise.} \end{cases}$$

Decision variable X_{iktj} is defined for demand points $i, j = 1, \dots, n$, $i \neq j$, and for hubs $k, t = 1, \dots, p$. If $k = t$ then X_{ikkj} represents the one-hub-stop service from demand point i to demand point j through hub k. In this case, cost of inter-hub flows is $ac_{kk}d(Q_k, Q_k) = 0$. When $k \neq t$, X_{iktj} represents the two-hub-stop service from i to j through hubs k and t, in that order. Mathematically, the problem can be stated as follows.

Problem P1

$$\text{Minimize} \quad \sum_i \sum_j O_{ij} W_{ij} c_{ij} d(P_i, P_j) X_{ij} + \sum_i \sum_k \sum_t \sum_j W_{ij}\big(a_1 c_{ik} d(P_i, Q_k) + a c_{kt} d(Q_k, Q_t) + a_2 c_{tj} d(Q_t, P_j)\big) X_{iktj} \tag{1}$$

$$\text{subject to} \quad O_{ij} X_{ij} + \sum_k \sum_t X_{iktj} = 1 \quad \text{for al } i, j, \tag{2}$$

$$Q_k \in E^2 \quad \text{for all } k,\ X_{ij},\ X_{iktj} \in [0, 1] \text{ and integer for all } i, k, t, j. \tag{3}$$

The first term in the objective function evaluates the cost of providing nonstop service for the routes in H. The second term evaluates the cost of transporting flows between demand points through hubs. It is given by the sum of costs of transporting units from demand point i (origin) to hub k, from hub k to hub t and from hub t to demand point j (destination). These costs are discounted using a_1, a and a_2 to reflect economies of scale.

In Problem P1, the level of interactions between any two hubs is determined by the two-stop routes they serve. The amount of flows channelled through hubs is sensitive to the hub locations and may be less than the total flow. This is due to the fact that the flows shipped nonstop from one demand point to another are not channelled through hubs. Also, since the flows from a demand point to other points are considered separately, a demand point may be served by different hubs on different routes. Further, when all routes originating or terminating at a demand point are in H, all flows to/from the demand point may be shipped nonstop. Thus, a demand point may not be served by any of the hub facilities.

In Problem P1, nonstop service is considered only for a predetermined set of routes H. Depending on the application, H may include all routes, making nonstop service a *possible* service alternative for every origin–destination pair. Recently, in Aykin (1991, 1994), we considered this problem in discrete solution space and referred to this policy as 'Nonstrict Hubbing Policy' since hub connection is not a requirement. In another limiting case, H may have no route; only hub connected services may be permitted between the (non-hub) demand points. Federal Express maintains a hub-and-spoke system operating under a similar policy. With the exception of certain city pairs, all shipments between the cities served by the system are channeled through the hubs. We also note that the models studied in O'Kelly (1986), Fotheringham and O'Kelly (1989) and Aykin and Brown (1992) are special cases of Problem P1 without nonstop services between demand points (i.e. $H = \emptyset$). An additional restriction in these studies requires all flows to/from a demand point be channeled through the same hub (single hub assignment).

3. Location-routing algorithm

Problem P1 involves both 0–1 variables (service types) and continuous variables (hub coordinates). Let the cardinality of H be $|H| = m$. Then the number of 0–1 variables is given by $p^2n(n-1)+m$ and the number of continuous variables by $2p$. A problem with $n = 50$ demand points, $p = 5$ hubs and $|H| = 300$, for example, has 61550 0–1 variables and 10 continuous variables. Thus, Problem P1 involves a large number of 0–1 variables and, with distance as a function of the hub coordinates, a nonlinear objective function. In most applications, solving Problem P1 using commercially available software or well known techniques of mathematical programming will be prohibitively time consuming, if not infeasible. Therefore, efficient heuristics exploiting the properties of the problem are needed.

Like the location-allocation problem involving non-interacting new facilities (Cooper, 1963, 1964), Problem P1 reduces into smaller subproblems if the hub locations or the service types are known. The problem decomposes into a number of shortest path problems involving service type decisions if the hub locations are available. And if the service types are known, then the problem reduces into a multifacility location problem. Thus, a solution to one subproblem can be used to simplify the problem into the other. The location-routing algorithm (LOCATION-ROUTING) solves these two subproblems separately in an iterative manner. This process is repeated until no further improvement in the solution is observed.

We now describe the subproblems and their solutions. If the hub locations, Q_k, $k = 1,\dots,p$, are known, then the problem reduces into the following integer program involving only service type decisions.

Problem P2

$$\text{Minimize} \quad \sum_i \sum_j W_{ij} C_{ij} O_{ij} X_{ij} + \sum_i \sum_k \sum_t \sum_j W_{ij} C_{iktj} X_{iktj} \tag{4}$$

$$\text{subject to} \quad O_{ij} X_{ij} + \sum_k \sum_t X_{iktj} = 1 \quad \text{for all } i, j, \tag{5}$$

$$X_{ij} \text{ and } X_{iktj} \in [0, 1] \text{ and integer for all } i, k, t, j, \tag{6}$$

where $C_{ij} = c_{ij} d(P_i, P_j)$ and $C_{iktj} = (a_1 c_{ik} d(P_i, Q_k) + a c_{kt} d(Q_k, Q_t) + a_2 c_{tj} d(Q_t, P_j))$. With the known hub locations, distances between the hubs and between the demand points and the hubs are known. Hence, C_{ij} and C_{iktj} can be easily calculated. A solution to P2 can be obtained by decomposing it into smaller shortest path problems, one for each route, and then choosing the lowest cost service types. That is:

$$X_{ij} = \begin{cases} 1 & \text{if } (i, j) \in H \text{ and } C_{ij} = \min\limits_{u,v} \{C_{ij}, C_{iuvj}\}, \\ 0 & \text{otherwise,} \end{cases} \tag{7a}$$

and

$$X_{iktj} = \begin{cases} 1 & \text{if } (i, j) \in H \text{ and } C_{iktj} = \min\limits_{u,v} \{C_{ij}, C_{iuvj}\}, \\ 1 & \text{if } (i, j) \notin H \text{ and } C_{iktj} = \min\limits_{u,v} \{C_{iuvj}\}, \\ 0 & \text{otherwise.} \end{cases} \tag{7b}$$

Given a solution to Problem P2 with the sets of nonstop routes $S_\infty = \{(i, j): X_{ij} = 1\}$ and hub connected routes $S_{kt} = \{(i, j): X_{iktj} = 1\}$, the objective function of Problem P1 can be rewritten as

$$\sum_{(i,j) \in S_{00}} W_{ij} c_{ij} d(P_i, P_j)$$
$$+ \sum_k \sum_t \sum_{(i,j) \in S_{kt}} W_{ij} \big(a_1 c_{ik} d(P_i, Q_k) + a c_{kt} d(Q_k, Q_t) + a_2 c_{tj} d(Q_t, P_j) \big). \tag{8}$$

The first term in (8) is constant since it involves only nonstop routes and distances between the demand points. By letting $D = \sum_{(i,j) \in S_{00}} W_{ij} c_{ij} d(P_i, P_j)$ and rewriting the second term in (8), Problem P1 is reduced into the following problem.

Problem P3

$$\text{Minimize} \quad \sum_k \sum_t V_{kt} d(Q_k, Q_t) + \sum_i \sum_k U_{ik} d(P_i, Q_k) + D, \tag{9}$$

where

$$V_{kt} = a \sum_{(i,j) \in S_{kt}} c_{kt} W_{ij} \text{ and } U_{ik} = a_1 \sum_t \sum_{(i,j) \in S_{kt}} c_{ik} W_{ij} + a_2 \sum_t \sum_{(j,i) \in S_{tk}} c_{ki} W_{ji}.$$

Problem P3 is a multifacility location problem with unknown hub locations $Q_k \in E^2$, $k = 1, \ldots, p$. But, unlike the well known multifacility location model, Problem P3 may involve asymmetrical flows between the hubs $V_{kt} \neq V_{tk}$ (e.g. when $W_{ij} \neq W_{ji}$ for some i, j), $k \neq t$, and a constant D in the objective function. To solve Problem P3, we rewrite the first term in objective function (9) as $\sum_{1 \le k < t \le p} (V_{kt} + V_{tk}) d(Q_k, Q_t)$.

The well known Weiszfeld (1937) type techniques can now be used to solve Problem P3. For the Euclidean distance, we use the following equation:

$$Q_k^{h+1} = \frac{\sum_{t \neq k} \frac{(V_{kt} + V_{tk})}{\| Q_k^h - Q_t^h \|} Q_t^h + \sum_i \frac{U_{ik}}{\| Q_k^h - P_i \|} P_i}{\sum_{t \neq k} \frac{(V_{kt} + V_{tk})}{\| Q_k^h - Q_t^h \|} + \sum_i \frac{U_{ik}}{\| Q_k^h - P_i \|}}, \quad k = 1, \ldots, p. \tag{10}$$

Starting with a set of initial hub locations, (10) is used iteratively to solve Problem P3 with the Euclidean distance $\| \cdot \|$. An optimal solution to this subproblem is guaranteed if the partial derivatives of the objective function do not become undefined during the iterations. This happens when two hubs or a hub and a demand point have the same location. To overcome this difficulty, we adopted the Hyperboloid Approximation Procedure (HAP) of Eyster, White and Wierwille (1973) and added a small positive constant ε to the Euclidean distance function $(\| \cdot \|^2 + \varepsilon)^{1/2}$. A summary of the algorithm is given below.

Step 0. Determine a set of initial hub locations $\{Q_1^0, \ldots, Q_p^0\}$. Set tolerances ν, $\varepsilon > 0$ and $h = 0$.
Step 1. Given the hub locations $\{Q_1^h, \ldots, Q_p^h\}$, solve Problem P2 using (7a,b) to determine the sets S_{00}^h and S_{kt}^h, for all k, t. If $h = 0$, go to Step 2. If $h > 0$ and $S_{00}^h = S_{00}^{(h-1)}$ and $S_{kt}^h = S_{kt}^{(h-1)}$ for all k, t, then stop.
Step 2. Using S_{00}^h and S_{kt}^h found in Step 1, reduce the problem to Problem P3. Set $r = 1$ and let $Q_k^r = Q_k^h$, $k = 1, \ldots, p$.
Step 3. Compute $Q_k^{(r+1)}$, $k = 1, \ldots, p$, using (10).
Step 4. If $\| Q_k^{(r+1)} - Q_k^r \| > \nu$ for some k, set $r = r + 1$ and go to Step 3. Otherwise set $Q_k^{(h+1)} = Q_k^{(r+1)}$, $k = 1, \ldots, p$, $h = h + 1$, and go to Step 1.

The procedures outlined above for the subproblems P2 and P3 are guaranteed to solve them optimally. The solution found by LOCATION-ROUTING algorithm (referred to as terminal solution), however, is not guaranteed to be a global optimal solution to Problem P1 since the hub locations and the route structure are not determined together. Starting from different initial solutions, this algorithm may converge to different terminal solutions.

4. Finding a starting solution

In addition to an enumeration approach, we considered four strategies for finding starting hub locations in the LOCATION-ROUTING algorithm. One of the strategies considered involves randomly generated hub locations on the plane E^2. The other three strategies and the enumeration approach choose starting hub locations from the set of demand points with the expectation that an optimal solution to Problem P1 and an optimal solution to the problem with a restriction on the hub locations limiting them to demand points should be close in terms of route structure and hub locations.

4.1. Randomly chosen points in E^2

With this approach, starting hub locations are chosen randomly in E^2. Since the hubs are facilities providing only service, they have no demand of their own unless they are located at some demand points. Therefore, depending on the initial locations, no route may be assigned to some hubs. When this happens, the LOCATION-ROUTING algorithm starts with one or more non-serving hubs. This initial nonconnectiveness may or may not persist during the search. To avoid this situation and explore the computational properties of the algorithm fully, starting hub locations were checked for route assignment and new locations were generated for the hubs with no initial assignment.

Since different terminal solutions may be obtained starting with different initial hub locations, the LOCATION-ROUTING algorithm is applied a number of times with different random starting solutions. The best terminal solution found is then taken as the solution of the problem. The solution found this way is expected to improve as the number of starting solutions tried increases. To facilitate the presentation of the results, we refer to this version of the LOCATION-ROUTING algorithm as CONTRP heuristic in the remainder of the paper.

4.2. Random demand points

Another approach considered involves choosing p demand points randomly as initial hub locations. The LOCATION-ROUTING algorithm is then used to obtain a terminal solution starting from these initial hub locations. Like in the CONTRP heuristic, several sets of demand points may be tried. The best terminal solution found is taken as the solution to Problem P1. Again, it is expected that the best terminal solution found will improve as the number of starting solutions evaluated increases. We refer to this version of the LOCATION-ROUTING algorithm as RDEMP heuristic.

4.3. Drop solution

Assuming a hub at every demand point, this approach starts with n ('open') hubs. Each hub is evaluated to determine the increase in the objective function value if it is closed. This is done by removing the hub locations from the set of s open hubs ($p < s \leq n$), one at a time, and solving Problem P2 with the remaining s-1 hubs. A solution to Problem P2 with s-1 hubs can be obtained easily from the solution with s hubs as follows; since the nonstop routes and the routes served by the hubs other than the one considered for closure in the solution with s hubs are not affected, they remain unchanged. Hence, the new solution with s-1 hubs can be developed by examining only the routes that are affected by the closure of the hub under consideration. The hub causing the smallest increase in the objective function value is closed and its location is removed from the set of open hubs. These steps are repeated with the remaining hubs until a solution with p hubs is obtained. The LOCATION-ROUTING algorithm takes these p demand point locations as the starting hub locations and searches for a solution to Problem P1 in E^2. In the remainder of the paper, we refer to this version of the LOCATION-ROUTING algorithm as DROP heuristic.

4.4. Drop & Interchange solution

The DROP approach provides only one starting solution. To investigate the possibility of obtaining better terminal solutions starting from improved initial solutions, we considered a drop and interchange approach. This approach improves the starting solution obtained by the DROP approach by interchanging hub locations with non-hub demand points. By moving a hub, say hub k, from its current location to a non-hub demand point, a new set of p hub locations is obtained. The new set differs from the current set of hub locations only in the location of hub k. This interchange is evaluated by solving Problem P2 with the new set of hub locations. If this results in an objective function value better than that for the current set, then the current set of hub locations is updated by moving hub k to its new location. To obtain a solution to Problem P2 with the new hub k location in an efficient manner, we use the solution of P2 with the current set of hub locations (current solution). To find the new service types for the routes served nonstop or by the hubs other than hub k in the current solution, the service types for these routes are compared only with the service types (both one-hub-stop and two-hub-stop) involving the new hub k location. All possible service types are considered for the routes served by hub k in the current solution. The interchange step is repeated until either no further improvement is observed or a limit on the number of interchanges is reached. In Aykin (1991), we tested this approach for Problem P1 in discrete solution space. Our results showed that this heuristic is very efficient in locating a good solution to the discrete problem.

As in the DROP heuristic, the demand points found in the interchange step are taken as the starting hub locations in the LOCATION-ROUTING algorithm and a search is performed for a solution to Problem P1 in E^2. We refer to this version of the algorithm as DROP&I heuristic. Both the DROP and DROP&I approaches provide only one starting solution. In these heuristics, the terminal solution obtained from this attempt is taken as the solution to Problem P1.

4.5. Enumeration

The enumeration heuristic considers all possible combinations of p demand points out of n. Each set of p demand points is taken as a starting solution in the LOCATION-ROUTING algorithm and a terminal solution is determined. The best terminal solution found is taken as the solution to Problem P1. Compared to the other approaches, a large number of starting solutions (C_p^n) are considered in the enumeration algorithm. Therefore, this approach is not feasible for large problems. Also, this algorithm does not guarantee an optimal solution to Problem P1 since it only enumerates the demand point configurations. The best terminal solution obtained with the enumeration algorithm, however, provides a bound on the best terminal solution that can be obtained by the RDEMP, DROP and DROP&I heuristics. Hence, it was used on a set of small problems to evaluate the performances of the proposed heuristics. To obtain an optimal solution to Problem P1, one needs to enumerate all route structures (i.e. all service type-route combinations) and solve Problem P3 to determine the best hub locations instead. There are $n(n-1)$ routes between n demand points. The number of possible service types for a route is p^2 when nonstop service is not permitted and p^2+1 when nonstop service is permitted. Hence, the number of different route structures for n demand points and p hubs is between $p^{2n(n-1)}$ and $(p^2+1)^{n(n-1)}$. This number is very large even for some small instances of the problem.

5. Computational results

The proposed heuristics were tested with two sets of randomly generated problems. The first set contained problems involving ($p=2$, $n=10$), ($p=2$, $n=15$), ($p=2$, $n=20$), ($p=3$, $n=10$), ($p=3$, $n=15$) and ($p=5$, $n=10$). With each problem in this set, we considered 2 sets of values for a, a_1 and a_2 parameters (quantifying the scale economies): i) $a=0.1$, 0.3, 0.5, 0.7, 0.9 with $a_1=a_2=a+0.10$ (i.e. the values 0.20, 0.40, 0.60, 0.80, 1.00) and ii) $a=0.1$, 0.3, 0.5, 0.7, 0.9 with $a_1=a_2=1.0$. Thus, each problem was solved with 10 different a-, a_1- and a_2-values. The second set of economy of scale values with $a_1=a_2=1.0$ (i.e. no scale economies between demand points and hubs) corresponds to the situation assumed in the studies on the hub location problems published in the literature. We formed 60 problem groups considering all combinations of 6 (p, n)-values and 10 a-, a_1- and a_2-values. In each problem group, 3 independent replications with the same problem parameters n, p, a, a_1 and a_2 were solved, resulting in a total of 180 random problems. Coordinates of demand points were generated randomly according to a uniform distribution over [0, 100] and the flows between them over [0, 50]. The unit transportation cost c_{ik} was assumed to be constant for all route segments. The heuristics were coded in FORTRAN and run in batch mode at roughly the same load on a VAX 8550 system.

All 180 problems were solved by the enumeration, CONTRP, RDEMP, DROP and DROP&I heuristics. Each problem was solved 10 times with the CONTRP and the RDEMP heuristics starting from different initial solutions. The stopping constant ν was initially set to 10^{-1} and subsequently reduced until a terminal value of 10^{-4} was assigned to it. With ν, the HAP constant ε was reduced from its initial value of 10^{-1} to its terminal value 10^{-7}.

To compare the solution obtained with a heuristic with the best known solution to a problem, we calculated the percentage difference as

$$\%\Delta = 100 \times \frac{\text{Solution found by heuristic} - \text{Best known solution}}{\text{Best know solution}}.$$

We also studied the effects of permitting/not permitting nonstop services between demand points. Two cases were considered: (i) 0% case in which nonstop services between (non-hub) demand points are not permitted and (ii) 100% case in which nonstop services are permitted between all demand points. Thus, the set H is either an empty set (0% case) or includes all routes (100% case). The same 180 problems were solved assuming 0% and 100% conditions. The minimum, average and maximum $\%\Delta$-values are reported in Tables 1a, 1b, 2a and 2b. Tables 1a and 1b summarize the results for the 100% case with two sets of economy of scale values and Tables 2a and 2b for the 0% case. Also included in these tables are the cumulative percentages of problems (referred to as cumulative $\%\Delta$ distribution) with solutions 0%, less than 0.1%, 0.5% and 1% difference ($\%\Delta$) in objective value.

The enumeration algorithm located the best solution in all problems in both the 100% and 0% cases. The results summarized in Tables 1a, 1b, 2a and 2b show that the average $\%\Delta$ is small for all four heuristics in both the 100% and the 0% cases with the largest being 0.48% (Table 2b). All four heuristics produced the same (best) solution in about 50% of the problems and solutions with less than 1% difference ($\%\Delta$) in the objective function value in more than 87% of the problems. The cumulative $\%\Delta$ distributions and the average and maximum $\%\Delta$ displayed in Tables 1a, 1b, 2a and 2b also show that the DROP&I heuristic in the 100% case and the CONTRP heuristic in the 0% case outperformed the other heuristics. The DROP&I heuristic located the best solution (with$\%\Delta = 0\%$) in 77.5% of the problems in the 100% case and the CONTRP heuristic in 82.5% of the problems in the 0% case. The proposed demand point based heuristics are based on the expectation that an optimal solution to Problem P1 and a solution to the same problem with a restriction on the hub locations limiting them to demand points should be close in terms of the route structure and hub locations. The differences observed in the performances of the CONTRP and DROP&I heuristics indicate that, for problems involving small number of hubs and/or small number of demand points, this may not be true if nonstop services are not permitted between (non-hub) demand points (i.e. in the 0% case). In this case, better solutions may be obtained with the CONTRP heuristic.

In the second part of our study, we considered a set of large problems involving $p = 3, 5, 10, 15, 20$ hubs and $n = 20, 30, 40, 60$ demand points. A total of eighteen problems (one replication in each group) were solved with the parameter values $a = 0.5$ and $a_1 = a_2 = 0.60$. The problem data were generated randomly as described before. The problems in this set were solved only with CONTRP, RDEMP, DROP and DROP&I. Since the solution time required by the enumeration algorithm increases drastically with problem size, it was not included in this part of the study. Again, each problem was solved by CONTRP and RDEMP 10 times. The results are summarized in Tables 3 and 4 for the 100% and the 0% cases, respectively. As seen from these results, the DROP&I heuristic outperformed the other three heuristics. This heuristic located the best solution in all but 3 problems in the 100% case and in all but 2 problems in the 0% case. The CONTRP and RDEMP heuristics performed at least as well as (and even better in 5 problems) the DROP&I heuristic in the problems involving small number of hubs

Table 1
Computation results with problem set I–100% case

a) scale economies: $a_1 = a_2 = a + 0.1$

Algorithm	%Δ			Cumulative %Δ distribution			
	Min	Aver	Max	0%	$\leq 0.1\%$	$\leq 0.5\%$	$\leq 1\%$
Enumeration	0.0	0.0	0.0	100	–	–	–
CONTRP	0.0	0.13	1.44	62	72	93	96
RDEMP	0.0	0.36	4.32	61	66	81	90
DROP	0.0	0.33	2.35	53	62	78	87
DROP&I	0.0	0.07	0.99	78	88	94	100

b) scale economies: $a_1 = a_2 = 1.0$

Algorithm	%Δ			Cumulative %Δ distribution			
	Min	Aver	Max	0%	$\leq 0.1\%$	$\leq 0.5\%$	$\leq 1\%$
Enumeration	0.0	0.0	0.0	100	–	–	–
CONTRP	0.0	0.19	2.82	59	70	88	96
RDEMP	0.0	0.42	8.95	61	69	88	93
DROP	0.0	0.18	2.25	56	68	89	97
DROP&I	0.0	0.04	0.48	77	88	100	100

($p \leq 5$). In a number of problems, the DROP&I heuristic located solutions better than the solutions obtained with the DROP heuristic. However, as seen in Table 4 for the problem involving ($p = 3$, $n = 60$), the starting solution located by the DROP heuristic led to a better terminal solution. This shows that a better starting solution is not always guaranteed to lead to a better terminal solution in the LOCATION-ROUTING algorithm. The results given in Tables 3 and 4 also show that the DROP

Table 2
Computational results with problem set I–0% case

a) Scale economies: $a_1 = a_2 = a + 0.1$

Algorithm	%Δ			Cumulative %Δ distribution			
	Min	Aver	Max	0%	$\leq 0.1\%$	$\leq 0.5\%$	$\leq 1\%$
Enumeration	0.0	0.0	0.0	100	–	–	–
CONTRP	0.0	0.10	2.16	79	87	94	97
RDEMP	0.0	0.34	4.32	67	73	80	87
DROP	0.0	0.38	3.20	47	57	76	88
DROP&I	0.0	0.21	2.47	66	78	87	93

b) Scale economies: $a_1 = a_2 = 1.0$

Algorithm	%Δ			Cumulative %Δ distribution			
	Min	Aver	Max	0%	$\leq 0.1\%$	$\leq 0.5\%$	$\leq 1\%$
Enumeration	0.0	0.0	0.0	100	–	–	–
CONTRP	0.0	0.06	1.76	86	90	94	99
RDEMP	0.0	0.48	9.41	78	82	87	89
DROP	0.0	0.37	4.69	57	67	80	90
DROP&I	0.0	0.23	3.19	62	73	86	94

Table 3
Computational results with large problems – 100% case ($a = 0.50$, $a_1 = a_2 = 0.60$)

p	n	%Δ			
		CONTRP	RDEMP	DROP	DROP&I
3	20	0.0	0.0	0.0	0.0
	30	0.0	0.0	0.0	0.0
	40	0.0	0.0	0.59	0.0
	60	0.0	0.0	0.48	0.48
5	20	0.13	1.21	0.0	0.0
	30	0.92	0.59	0.0	0.0
	40	0.0	0.76	0.67	0.60
	60	0.73	0.0	0.24	0.12
10	20	0.78	2.05	0.0	0.0
	30	0.62	2.44	0.22	0.0
	40	1.30	1.37	0.64	0.0
	60	0.43	0.91	0.41	0.0
15	30	0.88	1.62	0.0	0.0
	40	0.81	1.84	0.06	0.0
	60	0.46	1.84	0.06	0.0
20	30	0.73	1.43	0.0	0.0
	40	1.40	1.81	0.0	0.0
	60	0.78	1.54	0.0	0.0

heuristic performed well in problems with large number of hubs ($p \geq 10$). The maximum %Δ for these problems was 0.64%.

We now summarize our results with both sets of problems: All four heuristics are capable of locating good solutions to the problem. When compared with the solutions obtained by the enumeration heuristic in the first set of problems, the best solutions obtained with the four heuristics were on average no more

Table 4
Computational results with large problems – 0% case ($a = 0.50$, $a_1 = a_2 = 0.60$)

p	n	%Δ			
		CONTRP	RDEMP	DROP	DROP&I
3	20	0.0	0.0	1.21	1.21
	30	0.0	0.0	0.01	0.0
	40	0.0	0.0	0.0	0.0
	60	0.0	0.0	0.0	0.06
5	20	0.44	0.0	1.14	0.0
	30	0.10	0.0	1.11	0.0
	40	0.29	0.0	0.59	0.0
	60	0.31	0.0	0.31	0.0
10	20	0.80	2.86	0.37	0.0
	30	1.07	1.26	0.0	0.0
	40	1.13	2.32	0.43	0.0
	60	0.92	0.49	0.0	0.0
15	30	0.21	2.11	0.03	0.0
	40	1.11	2.61	0.0	0.0
	60	1.48	2.52	0.08	0.0
20	30	0.76	1.25	0.0	0.0
	40	0.65	2.00	0.07	0.0
	60	1.14	1.54	0.23	0.0

Table 5
Solution times for the large problems – 0% and 100% cases (min:sec)

p	n	CONTRP		RDEMP		DROP		DROP&I	
		0%	100%	0%	100%	0%	100%	0%	100%
3	20	1:12	2:41	1:11	1:14	0:10	0:22	0:11	0:10
						(0:07)	(0:14)	(0:09)	(0:09)
	30	3:50	3:57	3:54	3:12	1:17	1:00	1:18	2:23
						(0:56)	(0:56)	(1:11)	(2:18)
	40	5:59	9:50	5:54	7:55	4:57	5:14	5:22	4:34
						(4:01)	(4:00)	(4:27)	(4:23)
	60	13:12	26:49	14:18	18:27	32:40	34:21	34:46	33:11
						(31:44)	(31:21)	(34:01)	(32:09)
5	20	2:05	2:13	1:39	2:00	0:09	0:09	0:16	0:12
						(0:07)	(0:07)	(0:14)	(0:10)
	30	8:49	6:12	7:57	5:46	1:30	2:24	1:34	2:44
						(0:56)	(2:04)	(1:18)	(2:24)
	40	11:05	11:24	15:21	14:55	5:03	4:41	5:43	5:42
						(3:59)	(3:56)	(5:21)	(5:35)
	60	30:04	40:10	27:42	32:31	32:43	34:00	37:33	41:50
						(31:18)	(32:43)	(36:25)	(41:02)
10	20	6:31	4:13	1:58	1:49	0:10	0:09	0:27	0:17
						(0:06)	(0:06)	(0:21)	(0:14)
	30	11:16	11:03	12:11	11:23	1:05	1:17	2:55	2:58
						(0:56)	(0:55)	(2:47)	(2:49)
	40	23:45	28:10	23:57	24:08	4:29	4:24	10:38	10:10
						(4:05)	(3:55)	(10:13)	(9:56)
	60	82:30	69:00	76:01	63:52	39:41	32:00	66:33	77:49
						(30:47)	(30:06)	(60:22)	(76:44)
15	30	21:24	21:41	7:10	9:49	1:23	1:15	3:11	2:36
						(0:36)	(0:48)	(3:32)	(2:09)
	40	37:55	34:02	30:23	22:45	4:21	4:16	6:42	15:55
						(3:38)	(3:43)	(9:38)	(15:16)
	60	118:38	176:52	122:08	85:41	34:51	37:49	138:56	142:52
						(31:19)	(31:15)	(137:33)	(140:38)
20	30	45:30	46:46	7:21	11:47	1:00	0:57	1:59	1:59
						(0:38)	(0:39)	(1:37)	(1:37)
	40	57:01	60:14	33:00	36:56	4:24	4:25	18:55	11:15
						(3:29)	(3:18)	(18:15)	(10:20)
	60	123:22	133:22	117:31	105:29	33:08	31:57	211:16	87:26
						(30:24)	(29:28)	(208:44)	(84:57)

than 0.42% different in the 100% case and no more than 0.48% different in the 0% case. In terms of solution quality, DROP&I seems to be the best heuristic for solving Problem P1. The DROP heuristic also performed well in the large problems with $p \geq 10$. In some problems with small number of hubs ($p \leq 5$), the CONTRP and RDEMP heuristics performed better than the DROP and DROP&I heuristics.

Table 5 shows the solution times (excluding input and output) for the problems in the second set. We report the total time needed (in bold) for 10 runs with the CONTRP and RDEMP heuristics and the time needed to find a starting solution (within parentheses) together with the total time (in bold) for the DROP and DROP&I heuristics. The solution time with the CONTRP, RDEMP and DROP&I heuristics increases rapidly as n and/or p increase. As can be seen from Table 5, the CONTRP and RDEMP heuristics require more CPU time than the DROP&I heuristic in most problems. The results reported in Tables 1–4 and the solution times in Table 5 together suggest that conducting an initial search for a 'good' starting solution is clearly a better strategy on large problems than trying a large number of random starting solutions. A comparison of the results obtained with the DROP and DROP&I heuristics shows that the initial search for a starting solution, the most time consuming step in both heuristics, may be limited to a greedy type search. As seen in Tables 3 and 4, when the initial search was limited to a drop type search with the DROP heuristic, the terminal solutions obtained were very close (with small $\%\Delta$) or same with those obtained with the DROP&I heuristic. However, compared to the DROP heuristic, the DROP&I heuristic requires large amounts of CPU time in large problems with $p \geq 10$. Also, the CPU time needed with DROP&I increases as the number of hubs p and/or the number of demand points n increase. With the DROP heuristic, the CPU time increases only as the number of demand points n increases. These features make the DROP heuristic preferable to the DROP&I heuristic in large applications.

To examine the effects of scale economies on hub locations and route structure, we solved a problem involving 3 hubs and 15 demand points with parameter values $a = 0.1, 0.3, 0.5, 0.7, 0.9$ and $a_1 = a_2 = a + 0.1$. For this purpose, we used the enumeration algorithm and the CONTRP heuristic. With the enumeration algorithm, all $C_3^{15} = 455$ demand point combinations were considered as starting hub

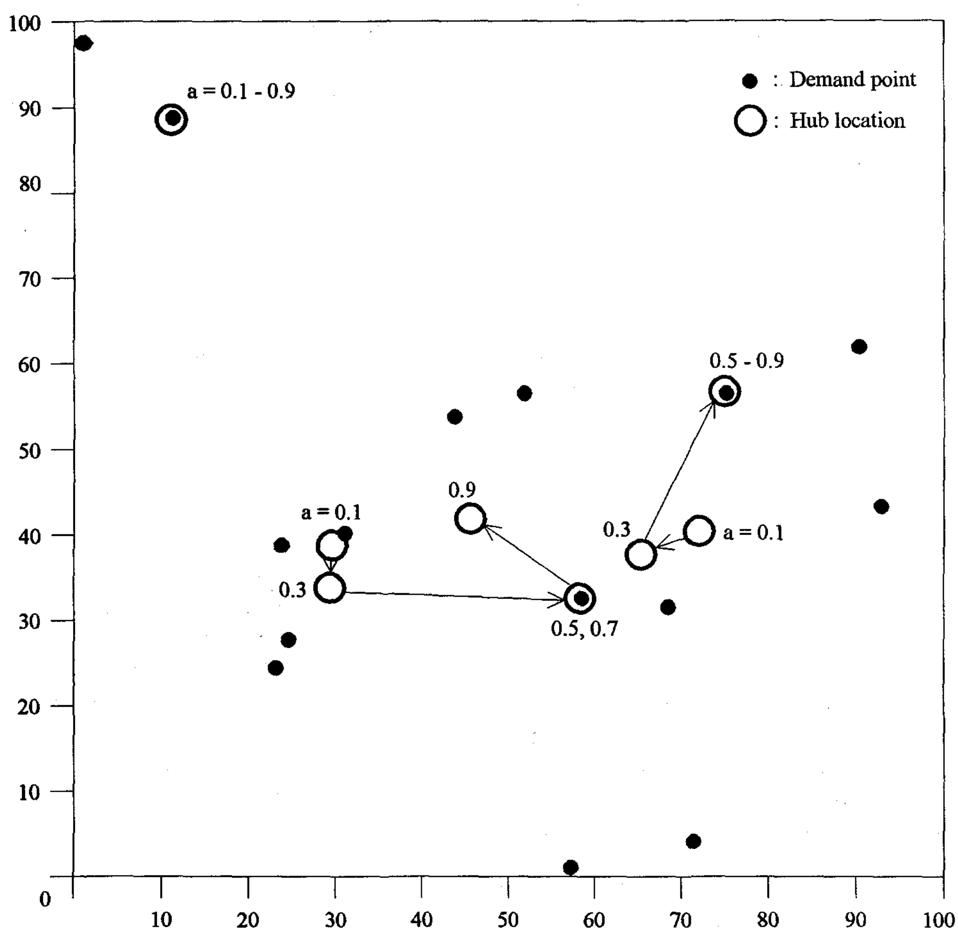

Fig. 2. 100% case with $n = 15$ and $p = 3$; demand points and hub locations ($a = 0.1, 0.3, 0.5, 0.7, 0.9$, $a_1 = a_2 = \alpha + 0.10$).

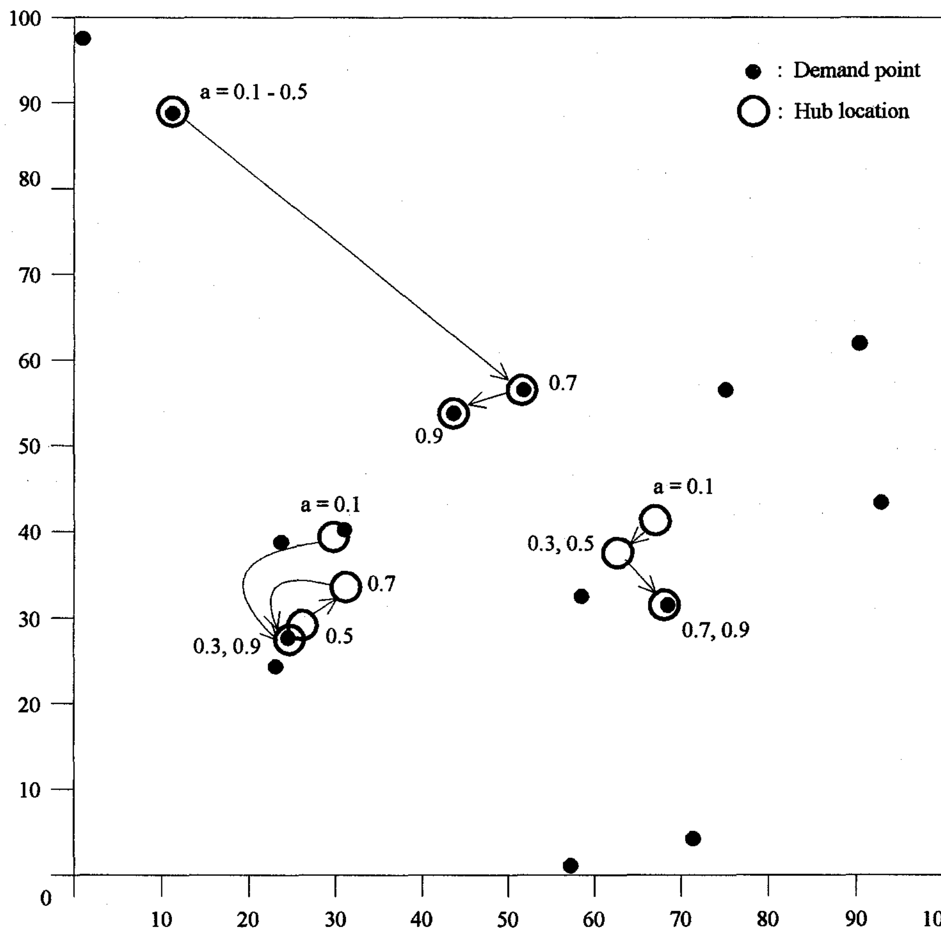

Fig. 3. 0% case with $n = 15$ and $p = 3$; demand points and hub locations ($a = 0.1, 0.3, 0.5, 0.7, 0.9$, $a_1 = a_2 = a + 0.10$).

locations. An equal number of randomly chosen initial solutions in E^2 were tried in the CONTRP heuristic. Thus, for each set of a-, a_1- and a_2-values, 910 terminal solutions were obtained. Hub locations in the best terminal solutions found are shown in Figs. 2 and 3 for the 100% and the 0% cases, respectively. As seen in these figures, the hub locations in the 100% and the 0% cases are about the same for $a = 0.1$, $a_1 = a_2 = 0.2$. Further, as seen in Figs. 4a and 4b, for these values of a, a_1 and a_2, the percentages of the routes served nonstop, one-stop and two-stop in the 100% and the 0% cases are approximately the same. Although in the 100% case nonstop service is permitted on all routes, only a very small percentage of the routes were nonstop (Fig. 4a). This is an expected result since small a-, a_1- and a_2-values make one-hub-stop and two-hub-stop services more attractive. Hence, the policy on nonstop service between demand points does not affect the service types and the hub locations significantly when the hub connected services are highly efficient and offer substantial scale economies. Consequently, similar networks are obtained in the 100% and the 0% cases. For large parameter values, however, networks with significantly different route structures and hub locations are obtained. In the 100% case, as a increases from 0.1 to 0.9 (and a_1, a_2 from 0.2 to 1.0), the percentage of nonstop routes increases from 2.9% to over 80% and the percentages of one-hub-stop and two-hub-stop service routes

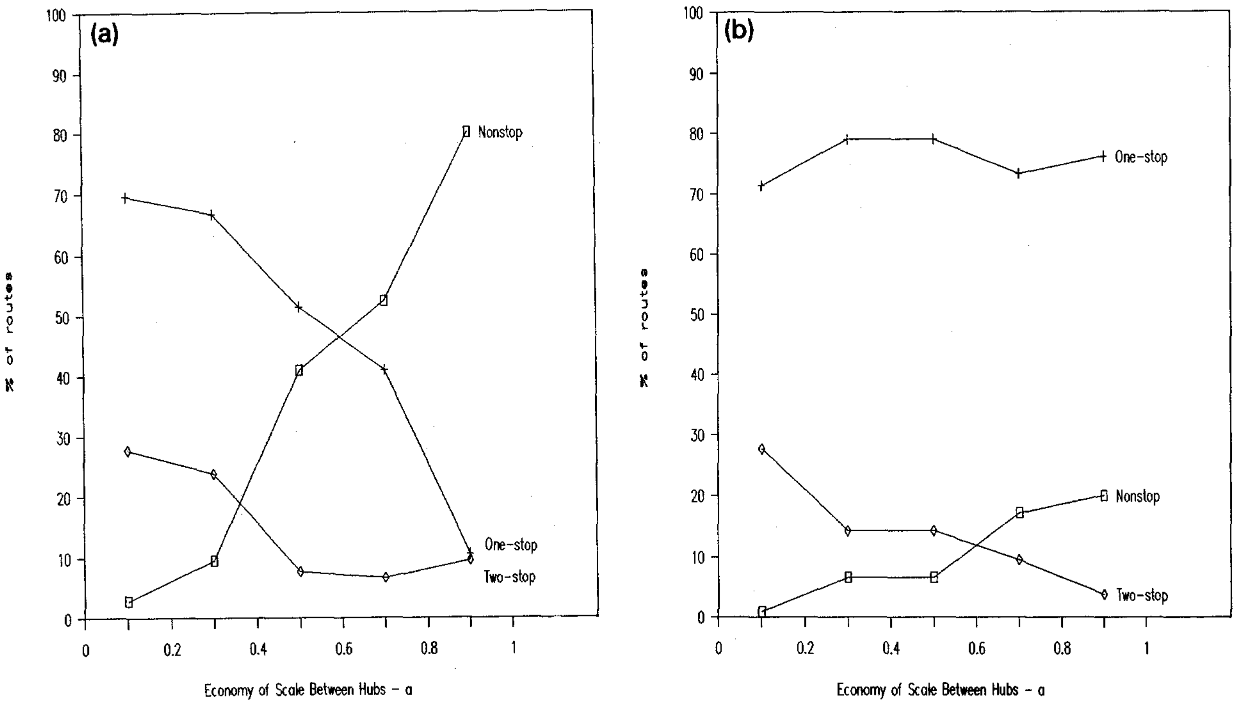

Fig. 4. Service types; a) 100% case and b) 0% case ($n = 15$, $p = 3$ and $a_1 = a_2 = a + 0.10$).

both drop below 10% (Fig. 4a). In the 0% case, on the other hand, the percentage of one-stop service routes remains high, between 71.4% and 79.1% (Fig. 4b). In the 0% case, nonstop service exists only if one or more hubs (e.g. all three hubs for $a = 0.9$, $a_1 = a_2 = 1.0$ in Fig. 3) are located at some demand points. When this happens, services between non-hub demand points and the demand points where the hubs are located, and between any two such hubs are considered to be nonstop. These results confirm that, depending on scale economies, the policy concerning nonstop services between demand points may be an important factor in the design of a hub-and-spoke system. Service types and hub locations that are optimal without nonstop services may become suboptimal when nonstop services are allowed between non-hub demand points.

The cumulative $\%\Delta$ distributions for this problem are shown in Figs. 5a and 5b for the 100% case and in Figs. 6a and 6b for the 0% case. Each cumulative distribution is based on 455 terminal solutions to the problem with the associated a (and $a_1 = a_2 = a + 0.1$) value. Figs. 5a and 6a are based on the terminal solutions obtained with the CONTRP heuristic and Figs. 5b and 6b on the terminal solutions obtained in the enumeration algorithm. Figs. 5a and 5b show that as a' increases from 0.1 to 0.9, the difference between the best and the worst terminal solutions in the 100% case decreases significantly. For $a = 0.9$ and $a_1 = a_2 = 1.0$, almost all of the terminal solutions are within 1% of the best solution. This indicates that, in the 100% case, 'good' terminal solutions (close to the best possible with a large number of starting solutions) can be expected in CONTRP and RDEMP with a small number of starting solutions when a, a_1 and a_2 are large. This can be explained by the service types; as seen in Fig. 4a, the percentage of nonstop routes increases to over 80% as a increases to 0.9. Although they do not affect the hub locations, service costs for nonstop routes are included in the total cost. Thus, the total costs for a terminal solution and for the best known solution to a case may not be significantly different even when

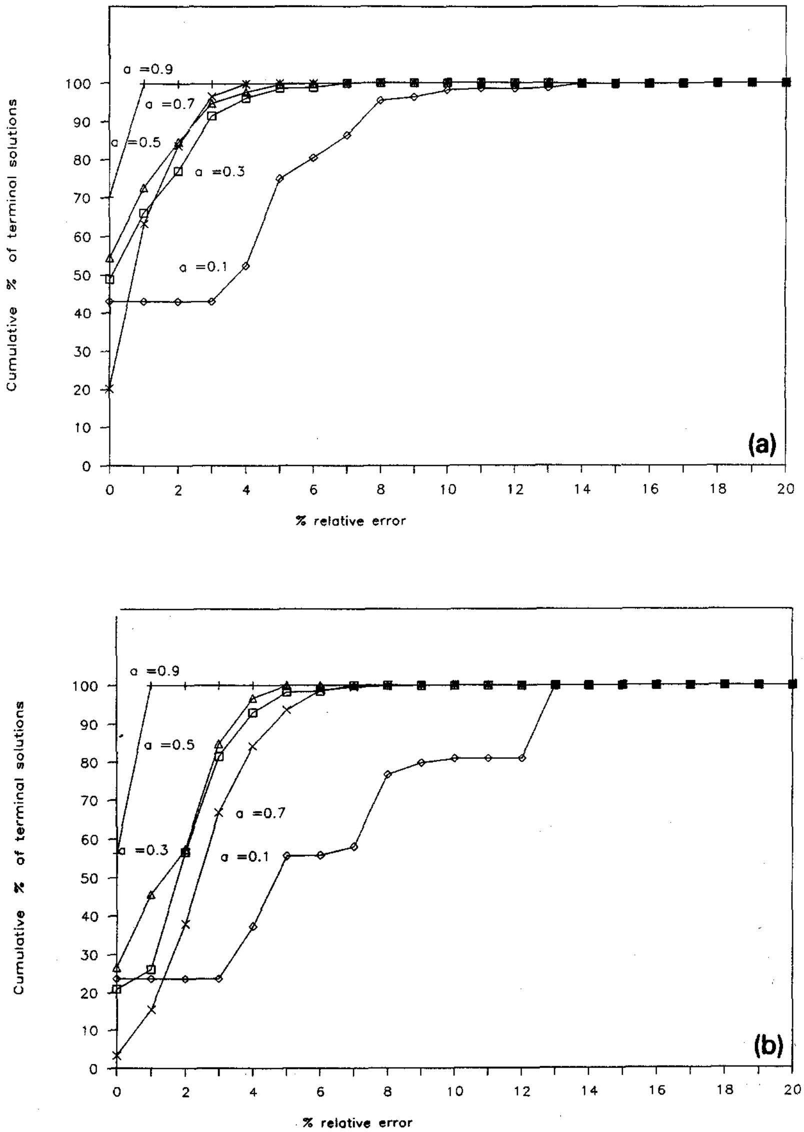

Fig. 5. Cumulative distribution of terminal solutions: 100% case ($n = 15$, $p = 3$ and $a_1 = a_2 = a + 0.10$). a) Random starting points in E^2 – CONTRP. b) Enumeration.

the hub locations and hub connected routes differ significantly. In contrast, for $a = 0.1$, less than 3% of the routes are nonstop. As seen in Figs. 5a and 5b, the total cost for a terminal solution in this case may differ from the total cost for the best known solution by as much as 14%. The cumulative $\%\Delta$ distribution in the 0% case (Figs. 6a and 6b) is not as sensitive to the changes in the economy of scale parameters as it is in the 100% case. For $a = 0.1$, all terminal solutions are within 16% of the best terminal solution and for $0.3 \le a \le 0.9$ within 10%.

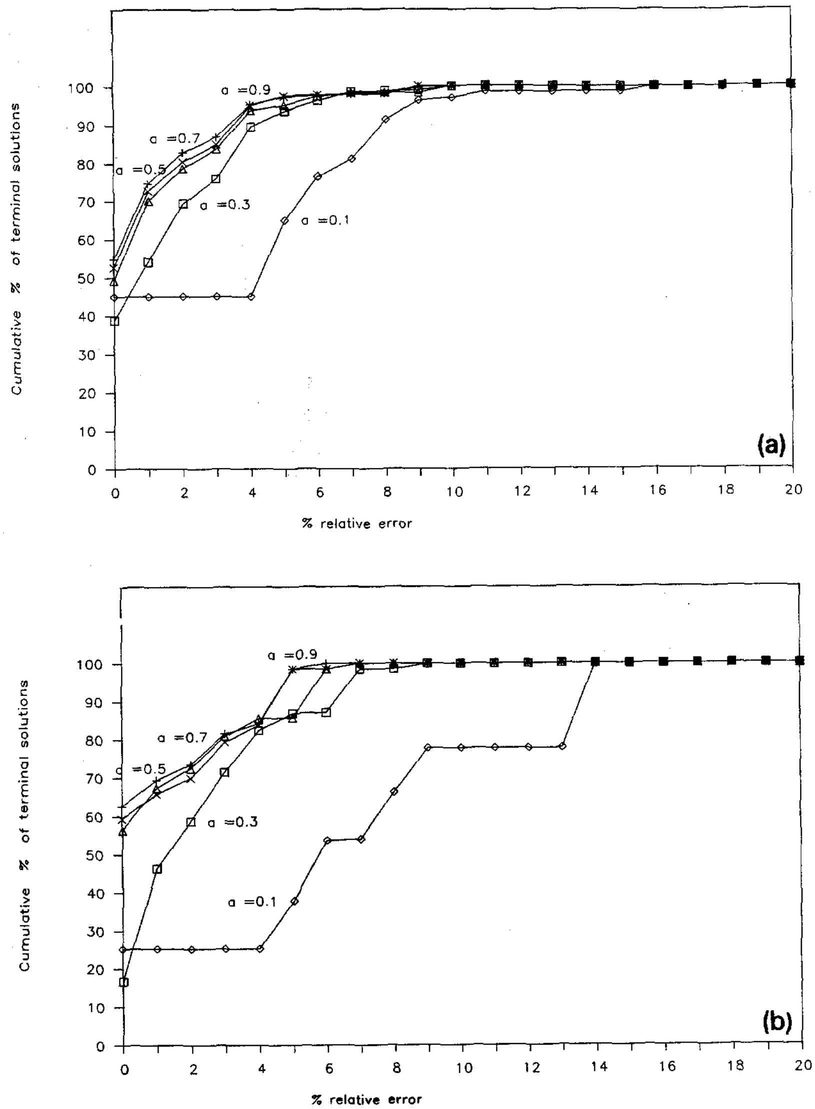

Fig. 6. Cumulative distribution of terminal solutions: 0% case ($n = 15$, $p = 3$ and $a_1 = a_2 = a + 0.10$). a) Random starting points in E^2 – CONTRP. b) Enumeration.

6. Summary

In this paper, we addressed the hub location and routing problem involving joint determination of the locations of interacting hubs and service types for the routes between demand points. In the situation considered, directional demand exists for the services between demand points. Flows from a demand point to other demand points are not aggregated but, instead, considered separately. Service provided from one demand point to another may involve channelling flows through one hub, two hubs or, if permitted, shipping nonstop. The hubs interact and the level of interaction between any two hubs is determined by the two-hub-stop service routes jointly served by them. We present a formulation of the problem on the plane E^2. To study the effects of the policy on nonstop services between (non-hub) demand points on the hub locations and route structure, we considered two cases: 100% case, permitting nonstop service between all demand points, and 0% case, not permitting nonstop service between demand points. We propose an algorithm solving a series of shortest path problems and a multifacility location problem in an iterative manner. Four heuristics based on the proposed algorithm were developed and tested. The proposed heuristics differ in the approach used to find starting solutions. We tested the proposed heuristics with two sets of random problems. The first set contained 180 problems and the second set included eighteen large problems. Among the four heuristics tested, the DROP&I heuristic is the best in terms of solution quality on large problems. This heuristic also performed well in small problems in the 100% case. In the problems involving small number of hubs and no nonstop service (0% case), CONTRP performed well. But when the CPU time needed is taken into consideration together with the solution quality, the DROP heuristic emerges as the most promising technique for large problems. Our analysis also demonstrated the importance of the policy concerning nonstop services between demand points in hub-and-spoke system design. Similar route structures and hub locations were obtained in the 100% and the 0% cases when significant scale economies exist (that is, small economy of scale constant values). But when the system does not exhibit significant scale economies, the route structure and the hub locations that are optimal without nonstop services may become suboptimal when nonstop services are permitted between demand points.

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